

Supplementary Material

A Low-Cost Single-Camera Adaptive Signal Controller for Heterogeneous Urban Traffic

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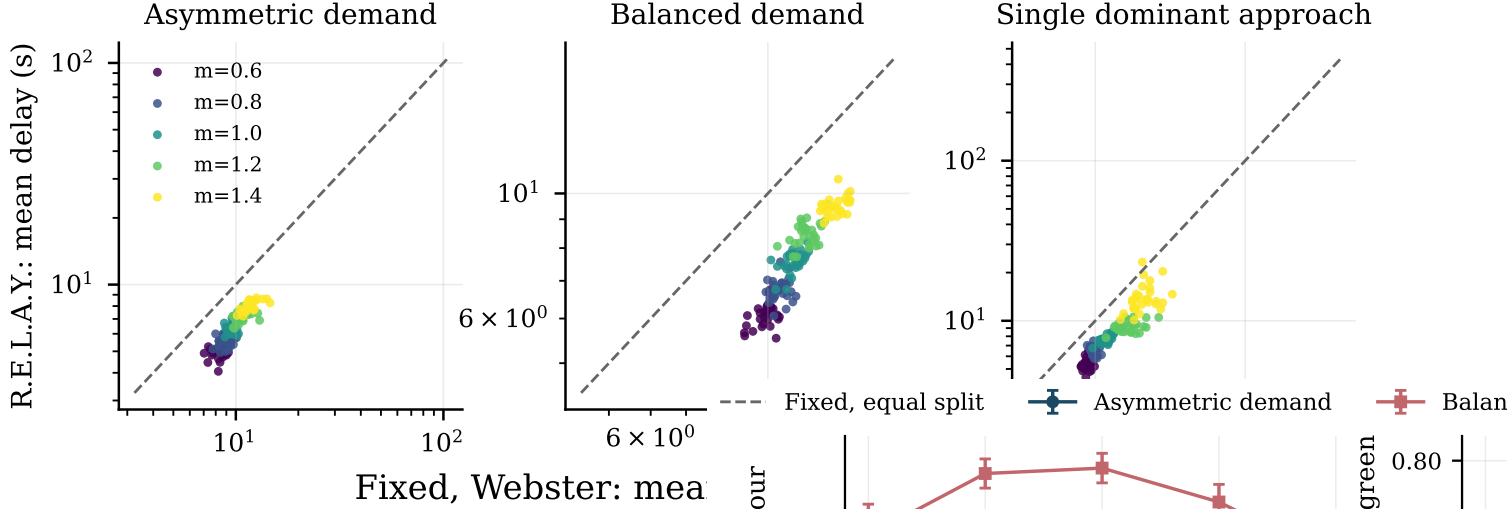


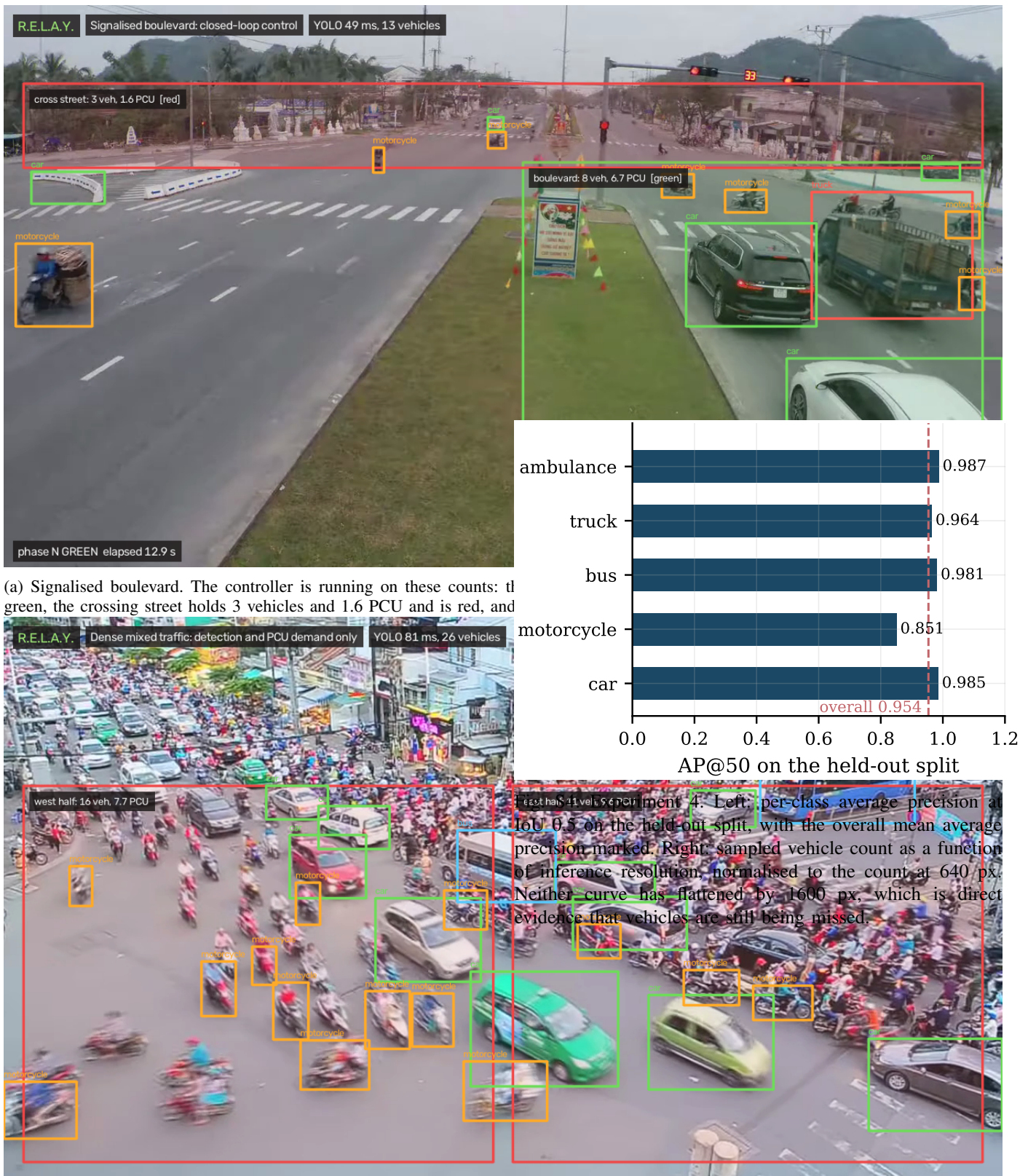
Fig. S1: Experiment 1, unaggregated. Each point is one seed at one demand level, plotting R.E.L.A.Y.'s mean delay against the Webster plan's mean delay on the identical arrival stream. Points below the diagonal favour R.E.L.A.Y.. Colour encodes the demand multiplier from 0.6 (dark) to 1.4 (light).

TABLE S1: Experiment 4. Detector performance on the system's held-out split: 97 images, 4990 labelled instances. The checkpoint evaluated is the one shipped with the system, a 2.59-million parameter model. P and R are precision and recall at the operating point the validator reports.

Model	Overall				AP@50 by class				
	mAP@50	mAP@50-95	P	R	car	motorcycle	bus	truck	ambulance
Shipped fine-tune	0.954	0.809	0.950	0.886	0.985	0.851	0.981	0.964	0.987

This supplement holds the unaggregated and per-class results referenced from the main manuscript. Nothing here is required to follow the argument; it is provided so every reported number can be inspected at the level it was measured.

Fig. S2: Experiment 1b. Left: phase changes per hour. Right: the fraction of wall time during which a green is displayed, the complement of which is time spent in yellow and all-red clearance. The dashed line is the equal-split fixed plan, whose behaviour is independent of demand by construction.



(b) Dense motorcycle-dominated traffic. The failure mode is visible: 26 vehicles are detected in a scene containing several hundred, so the reported PCU demand of 7.7 and 9.6 for the two halves is a severe underestimate. No controller decision is shown, because the junction depicted is not signal-controlled in the way the two zones imply.

Fig. S3: Experiment 4. The perception stack on real fixed-camera footage. Both frames are direct output of the system, with detections, approach polygons and per-approach PCU demand as computed at run time.